Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Deep Learning, NLP, Document Understanding | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

nishal.silva@hotmail.com | +1 613 200 2030 | Toronto, ON | nishal.xyz | linkedin.com/in/nishal-silva

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Re: Data Scientist, AI & Machine Learning – Scotiabank

Dear Hiring Manager,

I am applying for the Data Scientist, AI & Machine Learning role (requisition 274714) on Scotiabank's Global AI and Machine Learning team. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems from data ingestion through training, deployment and monitoring. Turning unstructured documents into reliable, production-ready data for the bank's business lines is the kind of problem I like: it needs careful evaluation as much as a good model.

Three parts of my work match your posting. For document understanding, at MAS Holdings I built OCR and layout-segmentation programs that fed document-processing bots to digitize garment tech-packs into a searchable archive, and I built computer vision inspection systems that cut inspection time by 99.5% and raised operational efficiency by up to 300%. For production ML, my published system runs at 14 ms on a Raspberry Pi 4 with F1 0.76, 74x faster than the DTW baseline (Journal of the Audio Engineering Society, 2026), and I owned it from data acquisition to deployment and monitoring in Python, Docker, Git and CI/CD. For data and risk work, at Forestpin I built ML and statistical pipelines in SQL and Python for financial forensics and compliance monitoring, working directly with compliance stakeholders to turn requirements into technical specs.

I should be direct about the gaps. I have not yet worked with Gemini in production; my LLM work is applied prompt design, RAG and HuggingFace tooling, and I expect to become effective with Gemini prompting for document extraction quickly. I have used Airflow, Snowflake and SQL on AWS rather than Hadoop or Spark, and I have not used Power BI. I have also worked in research settings more than in Agile delivery teams, though I have run code review and CI/CD practices with engineers.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the chance to discuss how my experience can support the Global AI and Machine Learning team at Scotiabank. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD